



2010 was a year of records setting for STMicroelectronics. The company clocked revenues of \$10.35 billion—the highest annual sales in its history. It was also a great year for the full range of its ‘sense and power’ portfolio (advanced analogue and MEMS, as well as microcontrollers and automotive applications).

Carlo Bozotti, president and chief executive officer of STMicroelectronics, was recently in India at the Greater Noida campus of the company to share the achievements. On the occasion, he spoke to Uma Bansal, executive editor at EFY, on semiconductor industry trends and ST’s contribution. Excerpts

■ CARLO BOZOTTI
PRESIDENT AND CHIEF EXECUTIVE OFFICER
STMicroelectronics

“We need to provide complete solutions, not just products”

Semiconductors are today at the heart of many exciting applications. What applications do you see as the most promising?

We are targeting four major application areas: Smart consumer devices (portable devices, home appliances, tablets to HDTV, etc), energy management and saving, healthcare and wellness, and trust and data security.

What are the key technology trends driving the semiconductor space globally?

From silicon technology point of view, there are two major blocks. One is continuation of Moore’s Law. It’s basically doubling the number of transistors per square millimetre every 18 months. This is still valid for all the VLSI technologies, and continues to be a strong and very important driver. Then there is the second block of technologies which we define as ‘more than Moore.’ Here the evolution is not

just in terms of the number of transistors or other things on the chip area but in terms of special functionalities like micro-machinery techniques and smart power techniques for power products, new materials like silicon carbide (SiC) and gallium nitride (GaN), or 3D integration.

Around these two blocks of technologies is the application aspect. An important trend we see today is that for more and more applications, we need to provide our customers complete solutions, not just products.

For portable devices, power consumption is very important. What’s the industry doing to reduce power consumption by chips?

We are continuously innovating in terms of technology to reduce the consumption of power in silicon and new materials like SiC. Also, there is a continuous drive to improve the topology or configuration of power applications to con-